ChemComm

COMMUNICATION

Published on 09 February 2026
Licence and permissions

Check for updates

Cite this: *Chem. Commun.*, 2026, **62**, 4843

Received 7th January 2026,
Accepted 6th February 2026

DOI: 10.1039/d6cc00114a

rsc.li/chemcomm

Chemoenzymatic synthesis of sulfated *O*-glycopeptides from human CD34

Yinyu Jiang,†[ab] Qi Du,†[c] Fengzhuo Wu,[b] Wei Zhang,[b] Yuan Ma,[a] Zhuojia Xu,[a] Zhonghua Li,*[c] Wenjing Ma*[ad] and Tiehai Li *[ab]

***O*-Glycosylation is a key modulator of CD34 biofunction. To investigate the distinct roles of different glycoform modifications, we report an efficient chemoenzymatic strategy to rapidly assemble a library of CD34 *O*-glycopeptides bearing sulfated and sialylated glycan variants, enabling the elucidation of how specific glycan modifications govern molecular recognition events.**

CD34, a highly glycosylated type I transmembrane glycoprotein, is widely expressed on hematopoietic stem and progenitor cells (HSPCs), vascular endothelial progenitor cells, and certain vascular endothelial cells.[1–3] It plays pivotal roles in a variety of physiological and pathological processes including cell adhesion, signal transduction, HSPC homing and engraftment, inflammatory responses, and immune-cell trafficking.[3–5] The extracellular sialomucin domain of CD34 is enriched with *O*-glycosylation modifications.[5] Variations in its glycan structures including sialylation, fucosylation, and sulfation directly determine the affinity and specificity of CD34 interactions with its ligands such as selectins (*e.g.*, L-selectin, P-selectin and E-selectin).[6–8] These modifications dynamically fine-tune the selectivity and strength of CD34-mediated cell adhesion, thereby enabling context-dependent functional regulation in both physiological and pathological settings.[3–5]

For instance, sialyl-Lewis *x* (sLe^x) motif is essential for high-avidity binding, which governs HSPC rolling and adhesion to the vascular endothelium during homing.[9] Furthermore, 6-sulfo-sLe^x on CD34 of high endothelial venules (HEVs) has been identified as a critical epitope for binding with L-selectin to mediate lymphocyte homing.[6,7] In addition, multidimensional binding assays have demonstrated that CD34 expressed on human HSPC-enriched cells binds to vascular E-selectin and P-selectin *via* sialylated and sulfated *O*-glycans, facilitating HSPC migration.[8] Despite these advances, the precise structural determinants governing selectin recognition remain incompletely understood, largely due to the synthetic challenge of accessing structurally well-defined sulfated and sialylated CD34 glycopeptides for systematic structure-activity studies.

To probe the molecular basis of how CD34 glycosylation governs selectin binding, we focus on a key 16-amino acid sequence (ELPTQGTFSNVSTNVS),[10] which is located in a heavily *O*-glycosylated region of the protein. Herein, we report an efficient chemoenzymatic strategy for rapidly constructing a library of CD34 *O*-glycopeptides that incorporate sulfated and sialylated glycan variants as well as their non-sulfated and non-sialylated counterparts (Fig. 1). Our approach begins with the chemical synthesis of a simple glycopeptide bearing a single α-*O*-linked GalNAc residue. This common intermediate is then diversified by a series of enzyme-catalyzed glycosylations.[11–13] The resulting library of structurally well-defined glycopeptides enables comprehensive elucidation of how specific glycan modifications direct molecular recognition with different selectins.

The chemical synthesis of structurally well-defined sulfated and sialylated glycopeptides remains a formidable challenge. A major hurdle is the incorporation of complex glycans onto the peptide backbone, accompanied by the request for stereoselective installation of sialic acid residue[14–17] and site-specific introduction of sulfonate group.[18,19] Furthermore, the final deprotection steps may encounter side reactions including β-elimination or the hydrolysis of the acid-labile sialic acid residues.[20–22] To overcome these limitations, we developed an efficient chemoenzymatic strategy by combining chemical preparation of a simple glycopeptide with an α-*O*-linked GalNAc residue and subsequent enzymatic diversification to install sulfated and sialylated glycan structures in a controlled manner. From chemically synthesized glycopeptide **1** (Scheme S1), robust glycosyltransferases and sulfotransferase were

[a] *State Key Laboratory of Chemical Biology, Shanghai Institute of Materia Medica, Chinese Academy of Sciences, Shanghai 201203, China. E-mail: mwj@xjtu.edu.cn, tiehaili@simm.ac.cn*
[b] *School of Chinese Materia Medica, Nanjing University of Chinese Medicine, Nanjing 210023, China*
[c] *Department of Biochemistry and Molecular Biology, School of Basic Medical Sciences, NHC Key Laboratory of Glycoconjugates Research, and Department of Hepatobiliary Oncology, Zhongshan Hospital, Fudan University, Shanghai 200032, China. E-mail: zhonghua_@fudan.edu.cn*
[d] *School of Pharmacy, Xi'an Jiaotong University, Shaanxi 710061, China*
† These authors contributed equally to this work.

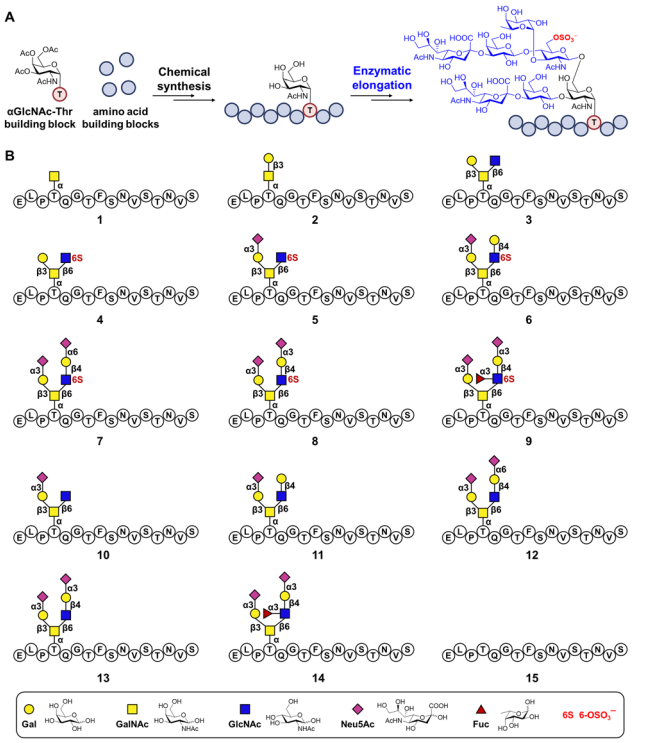
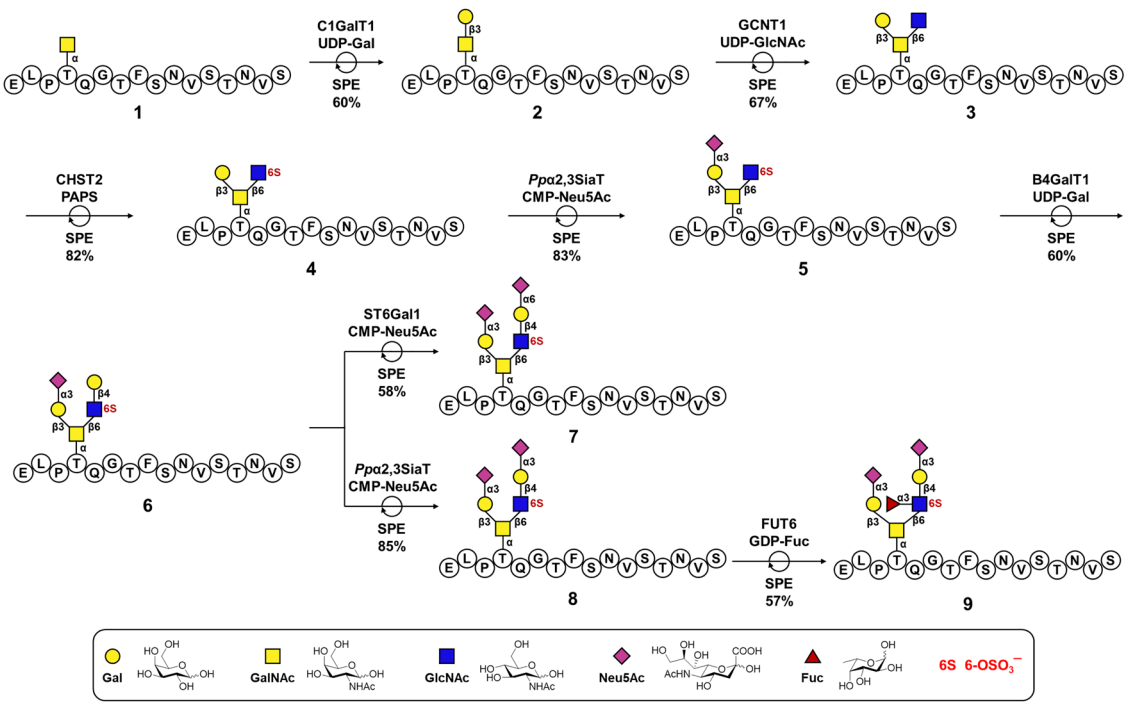


Fig. 1 Chemoenzymatic synthesis of diverse CD34 *O*-glycopeptides. (A) General strategy for chemoenzymatic assembly of complex *O*-glycopeptide. (B) Structures of the synthetic CD34 glycopeptide library.

used for the precise assembly of complex glycan chains on the glycopeptide scaffold (Scheme 1). First, the galactosylation of **1** was performed with β1,3-galactosyltransferase 1 (C1GalT1)[23] and UDP-Gal to provide glycopeptide **2**. The GlcNAc residue was then introduced to internal GalNAc moiety of **2** to afford the branched core 2 glycopeptide **3** using β-1,6-*N*-acetylglucosaminyltransferase 1 (GCNT1)[23,24] and UDP-GlcNAc. Subsequent installation of a 6-*O*-sulfate group on the GlcNAc residue of **3** was achieved using sulfotransferase CHST2[25,26] and 3′-phosphoadenosine-5′-phosphosulfate (PAPS) to provide glycopeptide **4** in a yield of 82%. The resulting glycopeptide **4** was sequentially elaborated by α2,3-sialyltransferase *Pp*α2,3SiaT[27] with CMP-Neu5Ac and β1,4-galactosyltransferase B4GalT1[28] with UDP-Gal to generate glycopeptides **5** and **6**, respectively. Next, divergent sialylation of the terminal Gal residue of **6** using ST6Gal1- and *Pp*α2,3SiaT-catalyzed reactions provided the corresponding disialylated glycopeptides **7** and **8**, which can be distinguished by their characteristic retention times on a HILIC column using LC-MS analysis (Fig. S17). Subsequently, based on the substrate specificity of α1,3-fucosyltransferase FUT6,[29] the terminal α2,3-sialylated LacNAc branch of glycopeptide **8** was selectively fucosylated to afford the complex CD34 glycopeptide **9** bearing sulfo-sialyl Lewis^x epitope. By applying the same enzymatic cascade to non-sulfated intermediate **3**, the corresponding non-sulfated glycopeptides **10-14** were synthesized in parallel (Fig. S1). During the process of enzymatic synthesis, intermediates and final products were rapidly purified by solid-phase extraction (SPE) using a C18 cartridge. All synthetic glycopeptides were characterized and confirmed by liquid chromatography-mass spectrometry (LC-MS) and high-resolution mass spectrometry (HRMS).

To enable high-throughput profiling of glycopeptide-protein interactions, a microarray was fabricated by printing synthetic core-2 glycopeptides **3-14** onto *N*-hydroxysuccinamide (NHS)



Scheme 1 Divergent enzymatic synthesis of a CD34 glycopeptide library featuring diverse sulfated and sialylated *O*-glycans.

 This journal is © The Royal Society of Chemistry 2026

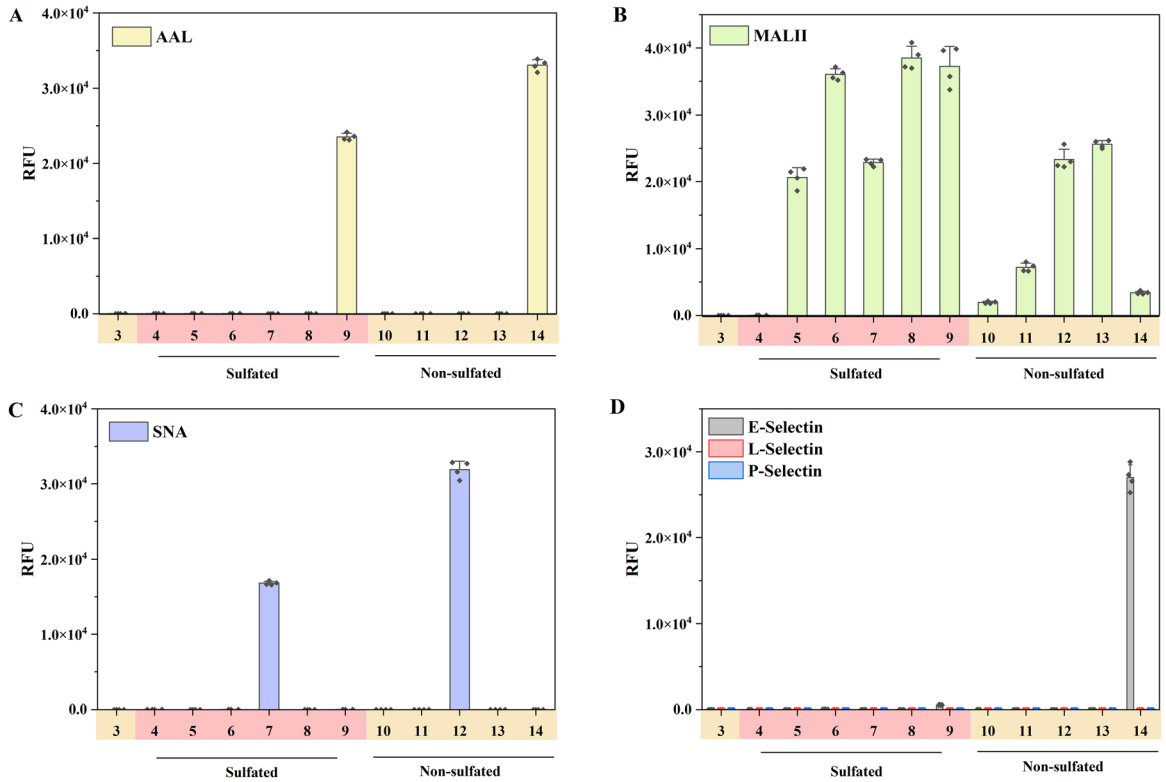


Fig. 2 Glycopeptide microarray binding analysis of synthetic glycopeptides (100 μM) with (A) AAL (10 μg mL^{-1}), (B) MAL-II (10 μg mL^{-1}), (C) SNA (10 μg mL^{-1}), and (D) selectins (50 μg mL^{-1}). The mean relative fluorescence unit (RFU) values are determined from four independent microarray replicates, excluding the lowest and highest signals. Standard deviation values are shown as error bars.

ester-activated glass slides to assess the influence of sulfation, sialylation and fucosylation on bindings.[30–33] First, the microarray was validated by a panel of plant lectins with known specificities: fucose-specific *Aleuria aurantia* lectin (AAL)[34] bound to glycopeptides **9** and **14**, α2,3-Neu5Ac-specific *Maackia amurensis* lectin II (MAL-II)[33,35] recognized glycopeptides **5-14**, and α2,6-Neu5Ac-specific *Sambucus nigra agglutinin* (SNA)[33,36] bound to glycopeptides **7** and **12** (Fig. 2A–C). The observed binding profiles matched the expected specificities, in agreement with prior reports. Next, the bindings of three selectins (E-, L-, and P-selectin) with CD34 glycopeptide library were investigated. Sialyl Lewis^x tetrasaccharide (sLe^x) motif has been reported as a common ligand for all three selectins,[9] while the specific influence of sulfation modification on these interactions remains unclear. Using our microarray containing both sulfated and non-sulfated CD34 glycopeptides, we assessed their binding profiles with each selectin. As shown in Fig. 2D, E-selectin exhibited strong binding to glycopeptide **14** bearing the sLe^x motif. In contrast, L- and P-selectin exhibited no obvious binding to any arrayed glycopeptide, including variants with sulfation and diverse α2,3- and α2,6-sialylation patterns. These results indicate that the sLe^x motif is necessary for E-selectin recognition under these conditions, and that sulfation alone or in combination with the sialylation patterns explored here is insufficient to confer binding to L- or P-selectin. It is suggested that efficient binding to L- or P-selectin may require the native, multivalent context of repeated and/or clustered sLe^x or sulfo-sLe^x motifs presented on the full-length CD34 glycoprotein.[37]

In summary, we have developed an efficient chemoenzymatic strategy that streamlines the assembly of a structurally well-defined library of CD34 *O*-glycopeptides bearing diverse sulfated and sialylated glycans. Microarray-based binding studies reveal that glycopeptides with sialyl Lewis^x tetrasaccharide motif is a key determinant for E-selectin recognition. This work enriches the synthesis space of complex glycopeptides and provides critical molecular-level insights into how specific glycan modification differentially regulates selectin-binding specificity, thereby advancing our understanding of glycan-mediated biological recognition.

The authors acknowledge the Strategic Priority Research Program of CAS (XDB1060000), the Shenzhen Medical Research Fund (B2402031), and the National Natural Science Foundation of China (22377134, 22507132, and 22307020).

Conflicts of interest

The authors declare no competing financial interest.

Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supplementary information:

supplementary Scheme S1, Figures S1–32, Tables S1–S2, and further experimental details. See DOI: https://doi.org/10.1039/d6cc00114a.

References

1 D. S. Krause, M. J. Fackler, C. I. Civin and W. S. May, *Blood*, 1996, **87**, 1–13.
2 M. Osawa, K.-i Hanada, H. Hamada and H. Nakauchi, *Science*, 1996, **273**, 242–245.
3 M. R. Hughes, D. Canals Hernaez, J. Cait, I. Refaeli, B. C. Lo, C. D. Roskelley and K. M. McNagny, *Exp. Hematol.*, 2020, **86**, 1–14.
4 L. Healy, G. May, K. Gale, F. Grosveld, M. Greaves and T. Enver, *Proc. Natl. Acad. Sci. U. S. A.*, 1995, **92**, 12240–12244.
5 C. R. Rodrigues, S. Moga, B. Singh and G. K. Aulakh, *Cell Tissue Res.*, 2023, **393**, 443–454.
6 K. Shailubhai, P. R. Streeter, C. E. Smith and G. S. Jacob, *Glycobiology*, 1997, **7**, 305–314.
7 G. H. Mir, J. Helin, K.-P. Skarp, R. D. Cummings, A. Mäkitie, R. Renkonen and A. Leppänen, *Blood*, 2009, **114**, 733–741.
8 D. B. AbuSamra, F. A. Aleisa, A. S. Al-Amoodi, H. M. Jalal Ahmed, C. J. Chin, A. F. Abuelela, P. Bergam, R. Sougrat and J. S. Merzaban, *Blood Adv.*, 2017, **1**, 2799–2816.
9 F. Jin and F. Wang, *Glycoconjugate J.*, 2020, **37**, 277–291.
10 A. Bister, T. Ibach, C. Haist, D. Smorra, K. Roellecke, M. Wagenmann, K. Scheckenbach, N. Gattermann, C. Wiek and H. Hanenberg, *Mol. Ther.: Oncol.*, 2021, **23**, 534–546.
11 R. M. Schmaltz, S. R. Hanson and C.-H. Wong, *Chem. Rev.*, 2011, **111**, 4259–4307.
12 C.-C. Liu, J. Ye and H. Cao, *Acc. Chem. Res.*, 2024, **57**, 636–647.
13 X. Chen, *Acc. Chem. Res.*, 2024, **57**, 234–246.
14 A. M. Vibhute, N. Komura, H.-N. Tanaka, A. Imamura and H. Ando, *Chem. Rec.*, 2021, **21**, 3194–3223.
15 P. Wang, S. Dong, J.-H. Shieh, E. Peguero, R. Hendrickson, M. A. S. Moore and S. J. Danishefsky, *Science*, 2013, **342**, 1357–1360.
16 B. Dhakal, A. Mandhapati, P. Eradi, S. Park, K. Fibben, K. Li, A. DeYong, S. Escopy, G. Karki, D. D. Park, C. A. Haller, E. Dai, L. Sun, W. A. Lam and E. L. Chaikof, *J. Am. Chem. Soc.*, 2024, **146**, 17414–17427.
17 Y. Li, X. Li, R. Wang, W. Liu, D. Liu, Z. Song, Y. Zhang, C. Zhang and Z. Tan, *J. Am. Chem. Soc.*, 2025, **147**, 17350–17360.
18 W. Yang, Y. Eken, J. Zhang, L. E. Cole, S. Ramadan, Y. Xu, Z. Zhang, J. Liu, A. K. Wilson and X. Huang, *Chem. Sci.*, 2020, **11**, 6393–6404.
19 T. Li, W. Yang, S. Ramadan and X. Huang, *Org. Lett.*, 2020, **22**, 6429–6433.
20 K. Baumann, D. Kowalczyk and H. Kunz, *Angew. Chem., Int. Ed.*, 2008, **47**, 3445–3449.
21 P. Sjölin, M. Elofsson and J. Kihlberg, *J. Org. Chem.*, 1996, **61**, 560–565.
22 N. Yamamoto, Y. Ohmori, T. Sakakibara, K. Sasaki, L. R. Juneja and Y. Kajihara, *Angew. Chem., Int. Ed.*, 2003, **42**, 2537–2540.
23 W. Ma, Y. Deng, Z. Xu, X. Liu, D. G. Chapla, K. W. Moremen, L. Wen and T. Li, *J. Am. Chem. Soc.*, 2022, **144**, 9057–9065.
24 T. Li, L. Liu, N. Wei, J.-Y. Yang, D. G. Chapla, K. W. Moremen and G.-J. Boons, *Nat. Chem.*, 2019, **11**, 229–236.
25 J. R. Grunwell and C. R. Bertozzi, *Biochemistry*, 2002, **41**, 13117–13126.
26 K. Huang, C. Li, G. Zong, S. K. Prabhu, D. G. Chapla, K. W. Moremen and L.-X. Wang, *Bioorg. Chem.*, 2022, **128**, 106070.
27 A. Mertsch, N. He, D. Yi, M. Kickstein and W.-D. Fessner, *Chem. – Eur. J.*, 2020, **26**, 11614–11624.
28 Z. Xu, Y. Deng, Z. Zhang, W. Ma, W. Li, L. Wen and T. Li, *J. Org. Chem.*, 2021, **86**, 10819–10828.
29 Y. Wu, G. P. Bosman, D. Chapla, C. Huang, K. W. Moremen, R. P. de Vries and G.-J. Boons, *J. Am. Chem. Soc.*, 2024, **146**, 9230–9240.
30 Z. Xu, Y. Liu, J. Liu, W. Ma, Z. Zhang, D. G. Chapla, L. Wen, K. W. Moremen, W. Yi and T. Li, *Nat. Chem.*, 2024, **16**, 881–892.
31 A. Y. Mehta, R. K. H. Veeraiah, S. Dutta, C. K. Goth, M. S. Hanes, C. Gao, K. Stavenhagen, R. Kardish, Y. Matsumoto, J. Heimburg-Molinaro, M. Boyce, N. L. B. Pohl and R. D. Cummings, *Cell Chem. Biol.*, 2020, **27**, 1207–1219.e1209.
32 S. Wang, C. Chen, M. R. Gadi, V. Saikam, D. Liu, H. Zhu, R. Bollag, K. Liu, X. Chen, F. Wang, P. G. Wang, P. Ling, W. Guan and L. Li, *Nat. Commun.*, 2021, **12**, 3573.
33 X. Song, H. Yu, X. Chen, Y. Lasanajak, M. M. Tappert, G. M. Air, V. K. Tiwari, H. Cao, H. A. Chokhawala, H. Zheng, R. D. Cummings and D. F. Smith, *J. Biol. Chem.*, 2011, **286**, 31610–31622.
34 P. R. Romano, A. Mackay, M. Vong, J. deSa, A. Lamontagne, M. A. Comunale, J. Hafner, T. Block, R. Lec and A. Mehta, *Biochem. Biophys. Res. Commun.*, 2011, **414**, 84–89.
35 A. Imberty, C. Gautier, J. Lescar, S. Pérez, L. Wyns and R. Loris, *J. Biol. Chem.*, 2000, **275**, 17541–17548.
36 N. Shibuya, I. J. Goldstein, W. F. Broekaert, M. Nsimba-Lubaki, B. Peeters and W. J. Peumans, *J. Biol. Chem.*, 1987, **262**, 1596–1601.
37 M. Cohen and A. Varki, *Int. Rev. Cell Mol. Biol.*, 2014, **308**, 75–125.

 This journal is © The Royal Society of Chemistry 2026